

Engineering Mathematics III (ENGE702)

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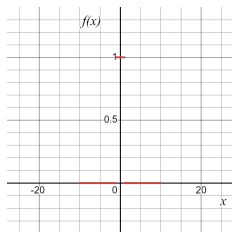
Lecture 20: Fourier Integrals and Transforms

Fourier Series and Integrals

We have analysed periodic functions using Fourier series. We can use similar methods (Fourier integrals and transforms) to analyse functions that are not periodic, and look at the whole x -axis.

We will use both these methods later as a tool to help solve PDEs.

For example, the function plotted below on the interval $[-10, 10]$ could be extended to be periodic on the whole x -axis or zero outside $[-10, 10]$.



We may approximate the periodic extension with a Fourier series and approximate the non-periodic extension with a Fourier integral.

Fourier Series

Example: Consider the family of functions (parameterised by $L > 1$), defined by:

$$f_L(x) = \begin{cases} 0 & \text{if } -L < x < -1 \\ 1 & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < L \end{cases}$$

and $f(x + 2L) = f(x)$. Then $f_L(x)$ will have the Fourier series coefficients:

$$a_0 = \frac{1}{2L} \int_{-1}^1 1 \, dx = \frac{1}{L}$$

$$a_n = \frac{1}{L} \int_{-1}^1 \cos\left(\frac{n\pi x}{L}\right) dx = \frac{2}{L} \frac{\sin\left(\frac{n\pi}{L}\right)}{\frac{n\pi}{L}} = \frac{2 \sin\left(\frac{n\pi}{L}\right)}{n\pi}$$

$$b_n = 0 \text{ as } f_L(x) \text{ is even.}$$

Fourier Series

Hence each function $f_L(x)$, $L > 1$, defined by:

$$f_L(x) = \begin{cases} 0 & \text{if } -L < x < -1 \\ 1 & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < L \end{cases}$$

and $f(x + 2L) = f(x)$, has the Fourier series:

$$f_L(x) = \frac{1}{L} + \sum_{n=1}^{\infty} \frac{2 \sin\left(\frac{n\pi}{L}\right)}{n\pi} \cos\left(\frac{n\pi x}{L}\right)$$

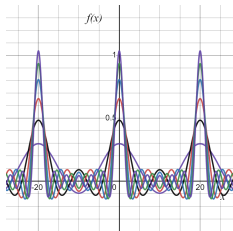
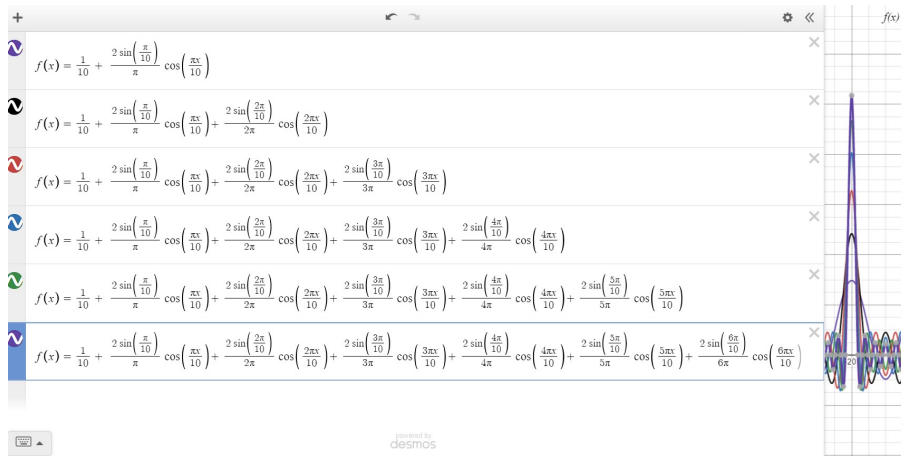


Figure: Successive approximations to $f_{10}(x)$ are periodic

Fourier Series



Fourier Series

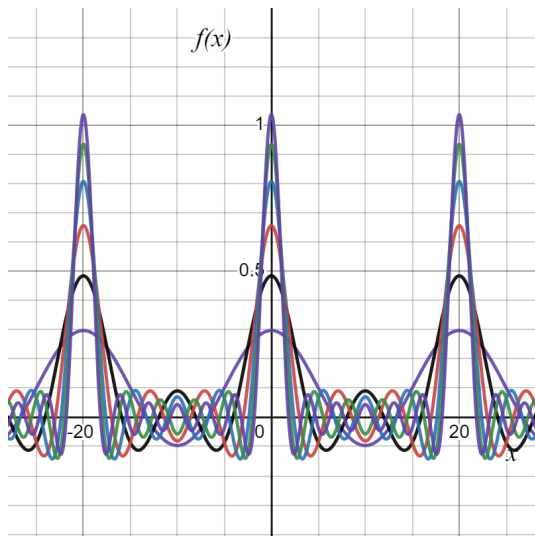
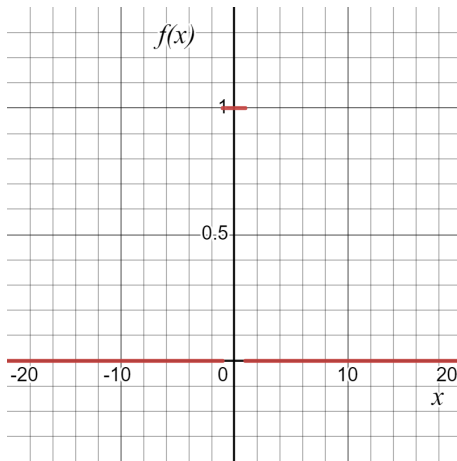


Figure: Successive approximations to $f_{10}(x)$ are periodic

Fourier Series



How do we approximate a function such as $f(x)$ where $f(x) = 1$ for $-1 \leq x \leq 1$, and $f(x) = 0$ otherwise? This function is not periodic.

Fourier Integrals

We only want one peak, around $x = 0$, not multiple peaks. We may take the limit as $L \rightarrow \infty$ (see textbook for details). Then our sum of sine and cosine terms at discrete frequencies $\frac{n\pi}{L}$ (our Fourier series) becomes an integral over all possible frequencies ω .

This becomes:

$$f(x) = \int_0^\infty (A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)) d\omega$$

where

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(\nu) \cos(\omega \nu) d\nu, \quad B(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(\nu) \sin(\omega \nu) d\nu.$$

There is a lot of theory behind these very useful formulae. You are encouraged to do some background reading. Start with the textbook.

Conditions

It is not hard to find functions $f(x)$ for which the integrals representing $A(\omega)$ and $B(\omega)$ do not exist (are infinite). Try $f(x) = \cos(\omega x)$. We need some conditions on $f(x)$ to ensure the existence of a Fourier integral. The following conditions are sufficient.

If $f(x)$ is piecewise continuous in every finite interval and has a right-hand derivative and a left-hand derivative at every point and if the integral

$$\int_{-\infty}^{\infty} |f(x)| dx$$

exists then f can be represented by a Fourier integral, that is,

$$f(x) = \int_0^{\infty} (A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)) d\omega$$

with functions A and B of frequency ω given on the previous slide. At a point where $f(x)$ is discontinuous, the value of the Fourier integral equals the average of the left-hand and right-hand limits of $f(x)$ at that point.

Conditions

Let's find a function $f(x)$ for which the integral

$$\int_{-\infty}^{\infty} |f(x)| \, dx$$

does not exist. Try $f(x) = \frac{1}{x^2}$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{x^2} \, dx &= \int_{-\infty}^0 \frac{1}{x^2} \, dx + \int_0^{\infty} \frac{1}{x^2} \, dx \\ &= \int_{-\infty}^{-1} \frac{1}{x^2} \, dx + \int_{-1}^0 \frac{1}{x^2} \, dx + \int_0^1 \frac{1}{x^2} \, dx + \int_1^{\infty} \frac{1}{x^2} \, dx \\ &= \lim_{t \rightarrow -\infty} \int_t^{-1} \frac{1}{x^2} \, dx + \lim_{t \rightarrow 0^-} \int_{-1}^t \frac{1}{x^2} \, dx \\ &\quad + \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x^2} \, dx + \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} \, dx \end{aligned}$$

Conditions

If any of the limits does not exist then

$$\int_{-\infty}^{\infty} \frac{1}{x^2} dx$$

does not exist.

$$\begin{aligned}\int_{-\infty}^{-1} \frac{1}{x^2} dx &= \lim_{t \rightarrow -\infty} \int_t^{-1} \frac{1}{x^2} dx \\ &= \lim_{t \rightarrow -\infty} \left(\frac{-1}{x} \right) \Big|_t^{-1} \\ &= 1\end{aligned}$$

but

$$\begin{aligned}\int_{-1}^0 \frac{1}{x^2} dx &= \lim_{t \rightarrow 0^-} \int_{-1}^t \frac{1}{x^2} dx \\ &= \lim_{t \rightarrow 0^-} \left(\frac{-1}{x} \right) \Big|_{-1}^t \\ &= \infty\end{aligned}$$

So $\int_{-\infty}^{\infty} \left| \frac{1}{x^2} \right| dx$ does not exist.

Fourier Integrals

Example: Find the Fourier integral for the following function:

$$f(x) = \begin{cases} 0 & \text{if } x < -1 \\ 1 & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x \end{cases}$$

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(\nu) \cos(\omega\nu) \, d\nu \\ &= \frac{1}{\pi} \int_{-1}^1 \cos(\omega\nu) \, d\nu \\ &= \frac{1}{\pi} \frac{\sin(\omega\nu)}{\omega} \Big|_{\nu=-1}^{\nu=1} \\ &= \frac{1}{\omega\pi} (\sin(\omega) - \sin(-\omega)) \\ &= \frac{2}{\omega\pi} \sin(\omega) \end{aligned}$$

Fourier Integrals

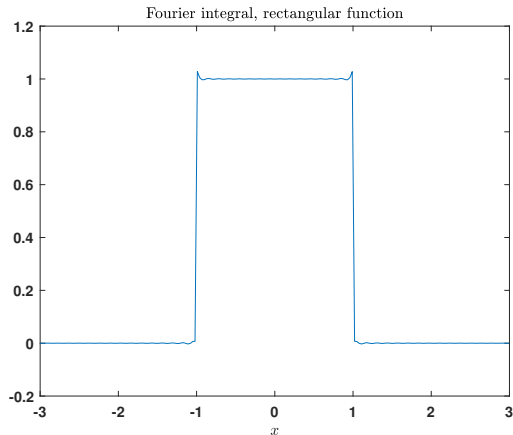
$$\begin{aligned} B(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(\nu) \sin(\omega\nu) \, d\nu \\ &= \frac{1}{\pi} \int_{-1}^1 \sin(\omega\nu) \, d\nu \\ &= \frac{-1}{\pi} \frac{\cos(\omega\nu)}{\omega} \Big|_{\nu=-1}^{\nu=1} \\ &= \frac{-1}{\omega\pi} (\cos(\omega) - \cos(-\omega)) \\ &= 0 \end{aligned}$$

$B(\omega) = 0$ since $f(x)$ is an even function

Hence,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin(\omega) \cos(\omega x)}{\omega} \, d\omega$$

Plot $f(x)$



In MATLAB, define a vector representing an interval on the x axis (use `linspace`) and calculate the integral at every point.

Fourier Transforms

Based on the integrals we have just done, we can define the Fourier transforms of a function $f(x)$ as follows:

$\hat{f}_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) dx$ is the cosine transform for an even function

$\hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(\omega x) dx$ is the sine transform for an odd function

We can then recover the original function by:

$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(\omega) \cos(\omega x) d\omega$ for an even function or

$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(\omega) \sin(\omega x) d\omega$ for an odd function.

Odd and even functions

Any function $f(x)$ is the sum of an even function and an odd function.
Why?

Let

$$g(x) = \frac{f(x) + f(-x)}{2}$$

and let

$$h(x) = \frac{f(x) - f(-x)}{2}$$

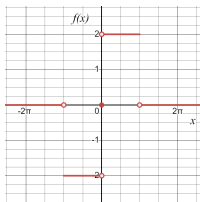
Then show $g(x)$ is even and show $h(x)$ is odd.

Then show $f(x) = g(x) + h(x)$.

Fourier Transforms

Example: Find the Fourier sine transform of the function:

$$f(x) = \begin{cases} 2 & \text{if } 0 < x \leq \pi \\ 0 & \text{if } \pi < x \end{cases}$$



Since we are asked for the sine transform, we extend the function as an odd function. We need to find $\hat{f}_s(\omega)$.

Fourier Sine Transform

$$\begin{aligned}\hat{f}_s(\omega) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(\omega x) \, dx \\&= \sqrt{\frac{2}{\pi}} \int_0^{\pi} 2 \sin(\omega x) \, dx \\&= 2 \sqrt{\frac{2}{\pi}} \left(\frac{-\cos(\omega x)}{\omega} \right) \Big|_{x=0}^{x=\pi} \\&= \sqrt{\frac{2}{\pi}} \frac{2}{\omega} (1 - \cos(\omega \pi)) \quad \text{so} \\f(x) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \frac{2}{\omega} (1 - \cos(\omega \pi)) \sin(\omega x) \, d\omega \\&= \frac{4}{\pi} \int_0^{\infty} \frac{(1 - \cos(\omega \pi))}{\omega} \sin(\omega x) \, d\omega\end{aligned}$$

Fourier Transforms

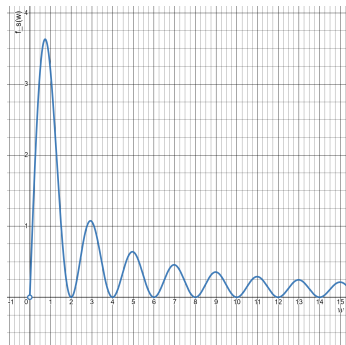


Figure: $\hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{2}{\omega} (1 - \cos(\omega\pi))$

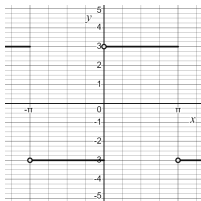
The Fourier sine transform shows the dominant frequencies. These are $\omega = 1, 3, 5, \dots$ with diminishing transform values proportional to $\frac{1}{\omega}$ or the square root of power. Compare the Fourier integral for $f(x)$ to the Fourier series obtained for a similar but periodic odd function in an earlier lecture. Use the Taylor expansion for $\cos(\omega\pi) = 1 - \frac{(\omega\pi)^2}{2!} + \frac{(\omega\pi)^4}{4!} - \dots$ to explore the $\lim_{\omega \rightarrow 0} \hat{f}_s(\omega)$.

For comparison

Earlier we found the Fourier series of the function:

$$f(x) = \begin{cases} -k & \text{if } -\pi < x \leq 0 \\ k & \text{if } 0 < x \leq \pi \end{cases}$$

and $f(x + 2\pi) = f(x)$.



namely

$$f(x) = \frac{4k}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)x)}{(2n+1)} = \frac{4k}{\pi} \left(\sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \frac{\sin(7x)}{7} + \dots \right)$$

For comparison

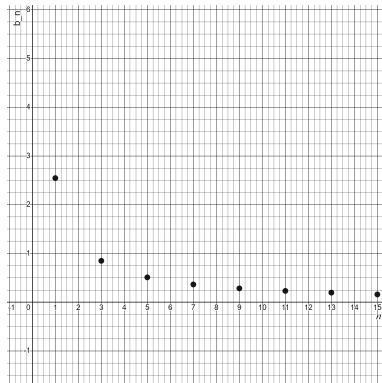
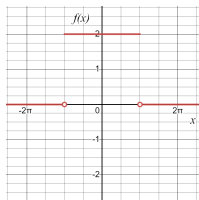


Figure: Discrete Fourier series coefficients for the periodic function in the previous slide, set $k = 2$, $b_n = \frac{8}{n\pi}$, compare with the continuous Fourier integral $\hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{2}{\omega} (1 - \cos(\omega\pi))$ for the previous non-periodic rectangular function

Exercise

- Find the Fourier cosine transform of the function:

$$f(x) = \begin{cases} 2 & \text{if } 0 < x \leq \pi \\ 0 & \text{if } \pi < x \end{cases}$$



Since we are asked for the cosine transform, we extend the function as an even function.

Draw the graph of $\hat{f}_c(\omega)$ and interpret.

Applications

Fourier analysis, spectral theory, harmonic analysis are used extensively, for example

- ▶ for solving PDEs
- ▶ in all kinds of signal processing (medical imaging, sound engineering, digital recording, radio communication)
- ▶ in Fourier transform infrared (FTIR) spectroscopy in chemistry, detecting compounds e.g. lipids and proteins (or contaminants such as melamine) in milk powder.
- ▶ The fast Fourier transform (fft) is an algorithm that computes the discrete Fourier transform of a sequence, for example converting sound in the form of a time series into information about the frequencies represented (see MATLAB livescript from the help pages: blue whale song)
- ▶ financial calculations (valuation of options)
- ▶ in quantum mechanics (wavefunctions)

Text Book References

Chapters 11.3, 11.7, 11.8